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Hox Activity Levels Governs the Evolution of Homologous Behaviors.

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Abstract

Central neuronal network rewiring is a key mechanism for the evolution of distinct behaviors. However, elucidating the genetic processes that have taken place to reconfigure circuits has rarely been performed. Here, taking advantage of the diversification of locomotor behaviors along the body of a single animal, we demonstrate how changes in the levels of the conserved developmental Hox genes govern the assembly of distinct circuits generating specific motor patterns. We show how these genetic changes occur during the late stages of development, refine the neuronal morphologies of a subset of neurons and lead to their incorporation into different circuits ultimately generating new function and behavior. Our work therefore uncovers a mechanism whereby evolutionary selected deviations in the expression levels of specification genes, refines connectivity in an otherwise unchanged network allowing for the evolution of a new behavior while ensuring system stability.

Main text

The extraordinary diversity of behaviors we observe today across or within species, has emerged through the evolution of the neural circuits that generate them. A mechanism that is beginning to be explored is the remodeling of central circuits through synaptic connectivity changes (aka rewiring) [1-6]. Working with simple nervous systems, such as the fruit fly or nematodes, has permitted identification of the neurons as well as the characterization of the synaptic changes that support distinct behavioral outputs. For example, the mating disinterest of *Drosophila simulans* males to females of the sister species *Drosophila melanogaster* correlates with a change in the excitatory and inhibitory balance onto a courtship promoting neuron, which is present in both species [1]. However, understanding the genetic changes, selected through evolution, that have induced circuit remodeling and shaped the diversification of function and behavior is still to be achieved.

The Hox genes are conserved master regulators of regional identity along the body in most animals, including humans. Evidence of their necessity for the diversification of body regions, both across species but also within the same species, have been well documented since the discovery of the development of an extra pair of wings in the *Drosophila Ubx* mutant [7,8]. In the vertebrate nervous system, we know that distinct patterns of Hox expression are necessary and sufficient for the assembly of the respiratory network [9,10]. A range of insightful studies have shown how the spatial and temporal patterns of Hox gene expression dictate the formation of along the spinal cord [11-14] and the appropriate connections with muscles [15-17]. Furthermore, distinct Hox patterning has been correlated with distinct locomotor outputs [18,19]. Therefore, the Hox genes are prime candidates to be driving central circuit rewiring to support the evolution of new behaviours.

In here, we used the locomotor behavior of the fruit fly *Drosophila melanogaster* larva as a model for behavioral evolution. The larvae explore their environment using two discrete behaviors: crawls and turns. Each one is autonomously generated by central circuits located at distinct levels along the ventral

nerve cord [20-22]. Left-right symmetrical waves of neuronal activity propagating anteriorly or posteriorly along the abdominal(a) nervous system (neuromeres) generate forward and backward crawls respectively. Forward and Backward crawls differ in the neuromere they originate from (a1 or a8) and also involve specific neurons that differentially control the wave propagation and the sequence of muscles contractions in each segment [23-27]. Turns, are generated by asymmetrical neuronal activity in the anterior segments, from thorax to a4 (Fig.1C) [21,28,29].

Despite the operational diversity between anterior and posterior abdominal segments, each neuromere arises from lineages of neuronal progenitors that are largely invariant [30-32]. Thus, this system offers the unique opportunity to investigate the role of neuronal circuit rewiring and its genetic control for the evolution of a novel locomotor behavior.

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Differences in neuronal morphology in a subset of interneurons underlie the specialization of function.

Given that the number and identity of interneurons is largely invariant across the abdominal segments a2-a7, with fewer neurons in a8 (Fig. 1a, b and Supplementary Fig. 1), we analyzed the regional differences in neuronal morphologies. Using extensive labelling of individual neurons from the Janelia collection [33] in third instar larvae with the multi-colour flip out method [34] we obtained over 20,000 images that we used to build an atlas (mini-atlas) of neural system diversity. We analysed the progeny of six neuronal progenitors, called Neuroblasts (NBs), that produce 43 of 270 abdominal interneuron types and are amenable to genetic targeting [35] (Fig. 1 and mini-atlas). Segment-specific morphological differences were defined by a series of conservative criteria. The presence/absence or change of direction of a primary dendritic or axonal branch was considered a change and was classified as posterior, anterior or commissural change. The length of a process was considered different only if it crossed to a new segment, whereas neurons whose dendritic shape changed in the same region of the neuromere were considered invariant (Fig. 1).

We found that the shape of more than half of the interneurons is invariant across abdominal segments (Figure 1a and c), suggesting that any diversity in the abdominal nervous system is incorporated into a basic segmental neuromere rather than each segment being completely reorganised.

Diversity in segmentally variant interneurons mostly arises from differences in the rostro-caudal direction and length of dendrites and axons or both, or in a few cases, in the differential presence of commissural projections (Fig. 2c; Supplementary Mini-atlas). The caudal shift of dendritic processes is the most common phenotype and consists of a gradual transition that occurs from anterior to posterior segments (Fig. 2b). For example, A23e (see method for neuron naming details) dendrites project mostly rostrally in a2. In a3 to a5, the dendritic tree extends rostral and caudally with branches of similar length but more lateral. In a6, the dendrites are mostly extending caudally (Fig. 1b). The dendritic shift could be accompanied by a shift in the axon in the same direction, as observed in A23e, or by no axonal change (i.e., A18b2 Supplementary Mini-atlas).

Other neurons showed an abrupt change in morphology as exemplified by the A18c and the A18g5/Canon neurons (when multiple names for an interneuron were used in the literature, we mention both once). Canon neurons, from abdominal segments 1 to 7, have an axonal projection that terminates

in a common region in thoracic segment 1 [24] suggesting a shared developmental programme. The homologous A18g5 interneuron in a8 has a short axonal projection terminating in a5 (Fig. 1b). The abrupt changes in morphology are often observed in neurons that are present in a1 (e.g., A18c) or a8 (e.g., canon).

Morphological analysis of homologous interneurons defines three diversification modes of neurons located in the VNC: i) neurons that are invariant regardless of the segment where they are located; ii) neurons with gradual changes of morphology along the antero-posterior neuraxis and iii) neurons that present abrupt changes in morphology.

A common feature of variable neurons is the conservation of the main skeleton (Supplemental Fig. 2) suggesting that their identity has been maintained but they may have responded to segment- and cell-specific developmental programs that have changed their fine wiring.

Hox expression deviates in a proportion of differentiating interneurons.

The antero-posterior differences in neuronal morphology suggested a role for the Hox genes, that are conserved master regulators of cellular identity along the body axis, in shaping wiring diversity [18,19]. Their effect needed to be not solely regional, but also generate cell specific differences. Hox genes are re-expressed in postmitotic neurons as they begin to differentiate (embryonic stage 16), put out axons and dendrites and form synaptic connections [37,38] suggesting they could drive the necessary network diversification. We analysed the levels of Hox proteins (ANTP, UBX, ABD-A and ABD-B) expressed in the abdominal segments in the NB lineages previously mentioned (Fig. 2a, b). The expression values normalised as a percentage of the highest expressing neuron in the same embryo [39].

The quantification revealed a wide heterogeneity in Hox combinations and levels (Fig. 2a). To understand the main characteristics of the Hox protein distribution, we used a kernel density estimation (KDE) analysis dividing the probability mass in five evenly spaced regions of 20% density (Fig. 2a and Supplementary Fig. 3). The KDE showed: first, that more than half (60%) of the neurons are expressing the generic Hox code expected for the segment. Second, that only a small proportion of neurons (20%) are expressing relatively high levels of one or more Hox protein (values above the last ring). Finally, it revealed peaks of maximum density (the most observed intensity, smallest ring) with very low levels of Hox expression (e.g. a2-a7 peak at 10% ABD-A and UBX).

We then wondered if the heterogeneity was only due to differences in UBX and ABD-A expression between the anterior and posterior compartment of each segment, as previously described [40-42] or, if it was present within lineages indicating a more complex regulation and possible role. We found the expected compartmental differences, with UBX highly expressed in the NB lineages located in the anterior half of the parasegment and ABD-A higher in NB lineages in the posterior half (Fig. 2B, fig S3).More interesting is our discovery that cells belonging to the same lineage expressed different profiles of Hox proteins (Fig. 2b-d, and Supplementary Fig. 4 and 5).This is evident in NB 2-4, where ABD-

A is absent or very low (<20%) in half of the interneurons from the lineage while the other half expresses an intermediate level of ABD-A (Fig. 1d) when located in a4, a canonical ABD-A region. Similarly, in a4, the neurons from NB 1-2 lineage express either high ABD-A or high UBX (Supplementary Fig. 4). Finally, we also found one neuron from NB5-6 that expressed UBX in a8, outside of the general pattern of expression between a2-a7 [41]. These cell-specific differences in the combinations of Hox proteins within NB lineages were present in all segments and in all NB lineages that we analysed (Supplementary Fig. 4 and 5).

Overall, our data show that for most neurons, the postmitotic Hox protein expression combination matches the generic parasegmental pattern described in the literature, but a proportion of individual neurons deviates considerably (Figure 1E). The fact that these deviations precisely occur during the process of differentiation, suggests that Hox proteins may play a role in the segment specific differentiation of equivalent cells along the antero-posterior axis.

Cell-autonomous Postmitotic Control of Neuronal Diversity.

To evaluate if the postmitotic expression of Hox genes drives the segment-specific differences in morphology, we focused on interneurons we could reliably identify in embryo and larva (Supplementary. Fig. 6). We worked with six interneurons from NB2-4 that could be recognised based on their morphology, birth order as determined with electron microscopy, and position within the lineage (Supplementary Fig. 6). Furthermore, two of these neurons, A18a and Canon, belong to the circuits generating crawling and they can be manipulated postmitotically [23,24] (Supplementary Fig. 7). Additionally, we included an interneuron involved in proprioception and required for left-right symmetry during crawling (A08e2) [43].

Invariant neurons have very low levels of Hox protein as in A18a (and A18b3, A18g3 A18c) and are insensitive to Hox over-expression as seen when over-expressing *abd-A, Ubx and ABD-B* in A18a (Fig. 3a and Supplementary Fig. 8). In contrast, differences in morphology correlate with segment specific higher levels of Hox protein suggesting the existence of a threshold which needs to be overcome for rewiring (Fig 3b,c, and Supplementary Fig. 8). A statistical comparison of the Hox levels in invariant versus variant neurons confirmed that significantly higher levels of Hox genes are present in variant neurons (Fig. 3d).

To empirically test the role of postmitotic Hox levels, we carried out gain of function experiments. We predicted that increasing Hox levels during differentiation will induce the transformation of morphology. In Canon neurons, ABD-B is relatively high in a8 and this correlates with a shorter axon. Over-expressing *Abd-B* in a1 and a6 resulted in much shorter axons, resembling the morphology of the homologous neuron in a8 (Fig. 3b',b"). This result indicates that ABD-B levels refine a specific axon developmental program in the terminal segment a8.

In A08e2, UBX levels are only high in a1 coinciding with anteriorly projecting axons and dendrites (Fig. 3c). When *Ubx* was over-expressed, we observed that the axons, which normally bifurcate into an anterior and posterior branch in segments a2 until a6, were uniquely projecting anteriorly as observed in

a1. *Ubx* over-expression did not produce a clear effect on the dendritic morphology of A08e2, only inducing an anterior lengthening of the lateral dendritic branch in a7 (Fig. 3b). Once more, we see that UBX levels modulate the development of the segmental A08e2 axonal direction of growth (Fig. 3b',b").

These results indicate that cell-autonomous postmitotic increase in Hox activity in specific differentiating neurons modulates specific developmental programs that determine the diversification of neuronal architecture.

Rewiring incorporates neurons into novel functional motor circuits.

We then asked to what extent Hox-driven changes in wiring affect their recruitment into motor circuits controlling specialized behavioral patterns. Working with newly hatched first instar larvae and using calcium imaging, we evaluated the patterns of activity of A18a and Canon neurons. Both neurons remained spontaneously active in isolated nervous system and are part of the circuitry for locomotion.

As a control for an invariant neuron, we analyzed the pattern of activity of A18a. Consistently with observations in a later larval stage *(20)*, A18a activity propagated in forward and backward waves, being part of the circuitry for forward and backward crawls (Fig. 4a). Over-expressing *Ubx*, *abd-A* or *Abd-B* in postmitotic A18a interneurons had no effect on their activity patterns (Fig. 4b,c and Supplementary Fig. 9). This observation is consistent with the lack of effect of this manipulation on the A18a morphology and suggests that Hox gene expression is not altering the physiological properties of the neuron.

We then performed the equivalent experiment analyzing the spontaneous activity pattern of Canon neurons, which are only active during fictive backward locomotion (Fig. 4d) and regulate the timing of muscle relaxation via excitation onto inhibitory pre-motor interneurons [24]. Over-expression of *Abd-B* in Canon neurons (called Canon>*Abd-B* from now on) induced a shortening of the axon, which is likely to affect the intersegmental connectivity between homologous neurons. The driver line used labels a variable number of Canon neurons, but when over-expressing *Abd-B* the number of neurons expressing calcium signals was decreased, suggesting they were inactive or absent (Fig. 4e and Supplementary Fig. 9). The active neurons showed different levels of intersegmental coordination. Some waves of activity propagated backwards along the nerve cord reaching a8 while in others, the pattern of activity was irregular within and across segments. We also observed the appearance of a greater proportion of forward waves of activity; a phenotype absent in the control line (Fig. 4d,f and [24]). This phenotype was specific to the over-expression of *Abd-B* and absent when over-expressing *Ubx* or *abd-A* (Fig. 4f and [24]).

To test if the forward waves on neuronal activity observed in Canon>*Abd-B* neurons are associated with fictive forward waves of locomotion, we recorded canon neurons simultaneously to aCC motorneurons. We found that a proportion of Canon neurons were active both during forward and backward fictive locomotion, while many neurons remained silent. This showed that the active neurons have become part of the forward and backward crawling circuit, while confirming that the development of a proportion of Canon > *Abd-B* neurons has been affected some to the extent of becoming inactive (Fig. 4g,h).

Finally, we wondered how the connectivity of Canon>*Abd-B* neurons might have changed to enable the integration into forward crawl circuit. On possible mechanism would be that the weight of their pre-synaptic connections to neurons which are active during forward and backward waves has changed [24]. We evaluated the identity of presynaptic partners and the location of synapses onto the axon of a wildtype a6 canon neurons, using reconstructions from an electron microscopy volume of a first instar larval central nervous system [24,44](supplementary Fig. 10). We found that the majority of the expected Canon-Canon synapses, necessary for the propagation of backward activity, occur along two to three segments away from the soma (Fig. 4i). The shortening of the axonal projection is likely to have affected these inputs, decreasing the drive during backward waves (Fig. 4j). Canon neurons also receive inputs from the brain descending backward locomotion command neurons MDNs [24,45]. MDNs synapses are mostly located in the more rostral axonal region of canon neurons, indicating that a shorter axon would probably diminish the strength of these inputs, affecting the backward propagation of Canon neurons activity in response to sensory stimulation. To become activated in forward waves, Canon>*Abd-B* neurons need to receive inputs from forward active neurons. In a6, the canon neuron on the right (not the left) receives one synapse from A27h, a neuron necessary for the propagation of forward crawl. This synaptic input is not sufficient to drive wildtype canon's activity during forward locomotion, but the close proximity between Canon and A27h neurons, that make contact several times at the dendrites close to the soma, suggests that Canon's rewiring could promote more or stronger inputs from the circuit involved in forward crawling. This combined with the predicted decrease of backward related inputs is likely responsible of the incorporation of canon>*Abd-B* neurons into forward pattern of activity.

Overall, these experiments show that rewiring specific neurons can have a dramatic impact on their connectivity and function allowing them to be incorporated into a novel specialized circuit.

Postmitotic Hox driven rewiring drives the evolution of behaviour.

It is difficult to imagine how the integration of rewired Canon > *Abd-B* interneurons into forward locomotion circuits, might drive circuit diversification. Wildtype Canon neurons induce segment relaxation after a backward peristaltic wave of contraction has passed. If the wave is travelling forward, the relaxation should occur one or two segment in front of the wave, in segments yet to be contracted (Fig. 3h). This might have no obvious behavioural effect or might affect the wave propagation. We found no difference in the speed of propagation of forward waves in canon>*AbdB* (Fig. 5a)suggesting there is no effect. We also quantified the time of backward wave propagation, that should be decreased if our prediction on Canon-Canon neurons connectivity weakening stands true (Fig. 5b and [24]). We did not find a significant effect but could observe a clear bimodal distribution of the larval behaviour, where half of the animals had their behaviour affected. This dichotomy was probably due to differences in the number of neurons affected in each individual as consequence of stochastic expression in the driver line (supplementary Fig. 7). Finally, we over-activated the neurons using thermogenetics (with UAS-*TRPA1*). In contrast to to the Canon > *TRPA1* that were almost completely paralyzed during the manipulation [24], many Canon > *TRPA1, AbdB* larvae performed forward waves (Fig. 5c). These escapers showed aberrant

phenotypes in that their peristaltic waves of muscle contraction progressed laboriously, indicating a behavioural impact of the rewiring induced by changes in the levels of *Abd-B* during differentiation. These results encouraged us to further test the role of rewiring in the VNC for the diversification of behaviour.

Many neurons show changes in morphology along the abdominal segments (Fig. 1). The additive effect of rewiring in several interneurons, that has taken place during evolution, is likely to have driven the establishment of distinct motor capacities for specific segments, allowing the diversification of homologous behaviours along the larval body [46-48]. To test this hypothesis, we evaluated the larval locomotor behaviour while over-expressing the *Ubx, abd-A* and *Abd-B* genes respectively in the thoracic and abdominal segments using the driver line *tsh*-gal4. A Gal80^{ts}-mediated conditional expression allowed us to perform this manipulation after NB proliferation has ended, during neuronal differentiation (Fig. 5d).

The larvae showed both losses and gains of phenotype as consequence of these manipulations. All larvae were able to execute coordinated crawls and turns (Fig. 5e-h) but they showed certain defects in locomotor performance ($\leq$ 2.5%) (Fig. 5i) such as: i) incomplete waves that stop before reaching the end of the body, ii) waves of contraction starting simultaneously in different segments (e.g a8 and a4) and iii) waves of contraction that changed direction before reaching the end of the body. Defective waves can be observed in all genotypes including control animals, but their frequency was slightly higher in genetically modified individuals. *tsh* > *Abd-B* presented a singular balance defect, their body shifting from left to right, as they steadily crawled (movie 1 and 2). The frequency and coordination of turns and head casts was unaltered (Fig. 5j). These phenotypes indicate that the establishment of segmental connections, that determine particular functions within the locomotor network, depends on Hox genes level of expression during differentiation.

Strikingly, we found two gain of function phenotypes. Over-expression of *Ubx* induced a significant increase of head lifting both during crawls and head casts, including turns (Fig. 5k-m and movie 3). Body lifting and lowering are necessary for displacement during peristaltic forward crawling as they allow the contracted segment to move forward, without being dragged over the substrate. *Ubx* mutants have defective lifting and lowering of segments a1 to a3 [49]. Therefore, we see that rewiring, occurring during differentiation, has been sufficient to drive the acquisition of a novel behaviour, fundamental for the proper execution of forward crawling, in thoracic segments, suggesting a transformation towards forward waves.

The second phenotype was a significant increase in the number of backward waves upon the over-activation of *abd-A* (Fig. 5f and movie 4). The phenotype consists of the spontaneous initiation of bouts of backward waves, interrupted by head cast, as occurs during forward crawling. The anterior initiation cite suggests a reorganization of connections such that thoracic segments have become more similar to abdominal segments. The changes are likely to include motor and proprioception circuits that drive the initiation and propagation of peristaltic crawling waves.

In summary, locomotion was not impaired by the changes in Hox genes levels of expression during differentiation, but the neuronal circuits were remodeled in ways that allowed the emergence of new behaviours in homologous segments.

Discussion

Together our results support the idea that the evolution of a new homologous behaviour is determined by the rewiring of a subset of interneurons. Differences in morphology change the synaptic connectivity between homologous neurons, allowing cells to meet new or more partners or modifying the synaptic weight between current synaptic partners. These differences have the potential of reconfiguring the patterns of activity between neurons supporting the evolution of new circuits, or circuit motifs, for specialized behaviours (Fig. 5n). In this scenario, region-specific changes of morphology of a subset of interneurons within a conserved network constitute a mechanism by which a new behavioral program can evolve while retaining the main properties of the network [6].

From a genetic point of view, the rewiring mechanism underlying network evolution results in low impact for the general structure of the central nervous system or low pleiotropy [6] . Our experiments show that specialization of morphology does not completely reconfigure the topology of the neurons. Although these neurons show diversity in presence, length, or orientation of their processes, they retained their main skeleton, which is an identity feature. Thus, only the late program of differentiation has been modified in a cell-autonomous manner in a subset of neurons, while retaining the main developmental program intact. We propose that modifications occur only in the final stages of development, which has the advantage of sparing the nervous system from wider complexity circuit reorganization that may generate deleterious consequences.

In the motor system, we found that cell-specific deviations in postmitotic Hox protein levels determine the diversification of morphology during differentiation. It is likely that they modify the expression of effector proteins (or also called realizators) [50-52] that control axonal and dendritic trajectories [53], synaptic organisation, and connectivity [15,38,54,55]. There is no apparent rule for the way Hox genes control the neuronal morphology, but rather a cell-specific effect where each cell slightly tunes their endogenous developmental program differentially in response to distinct levels of Hox genes [51]. Notably in the cortex, several transcription factors expressed postmitotically contribute to the establishment of subtype-specific morphology features [56]. Therefore, we see a conserved mechanism, where specification genes refine the neuronal architecture and connectivity allowing for further functional diversity [56-58], while retaining the main developmental program.

Our work thus uncovers a mechanism by which evolutionary selected fortuitous deviations in specification genes levels rewire a subset of neurons in ways that drive their incorporation into a novel specialized circuit. Such a mechanism ensures the system stability and simplifies circuit diversification—within organisms and potentially also across homologous behaviour in all animals (Fig. 5n).

Declarations

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Data and materials availability:

Mini-atlas will be uploaded on webpage, it is available on request.

KDE was performed in Julia programming language using the MarginalHist function from StatsPlots.jl (https://github.com/JuliaPlots/StatsPlots.jl), which utilises KernelDensity.jl (https://github.com/JuliaStats/KernelDensity.jl).

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Figures

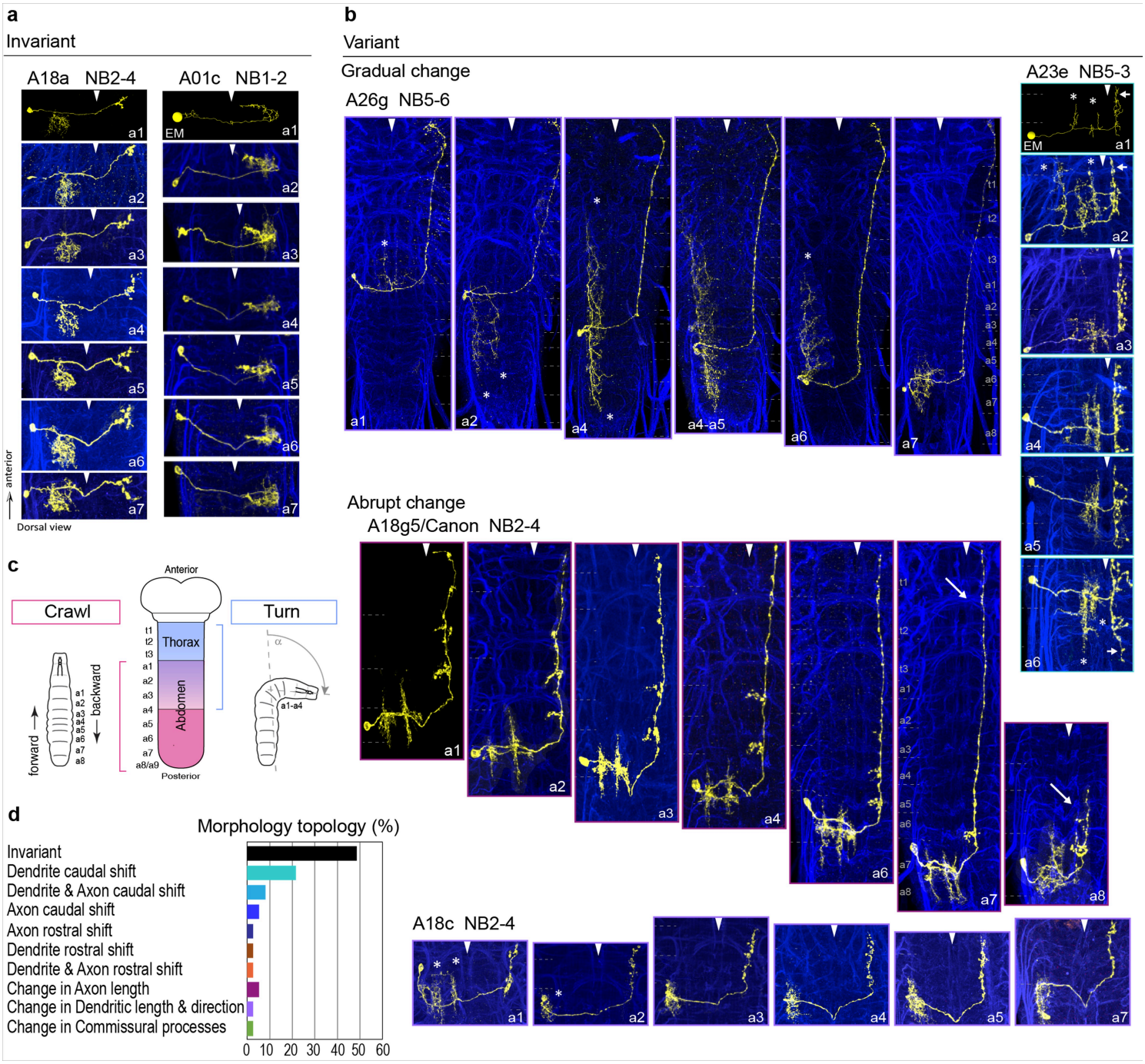


Figure 1

Diversification of neuronal morphologies is present in a subset of abdominal interneurons.

a. Example of two interneurons with invariant morphologies from neuroblasts (NB) 1-2 and 2-4. The segment where the soma is located is indicated in the corner of each image. Color of flip-outs was homogenized with ImageJ. Top arrowhead indicates the midline; EM that the image was obtained from Electron Microscopy tracing.

b. Example of interneurons with variant segmental morphologies. Segment differences in dendrites and axons are indicated by asterisks or arrows, respectively. *Gradual change:* The interneurons A26g and

A23e show gradual changes in their processes along the antero-posterior axis. A26g is a projection neuron with dendrites that change direction and length along the abdominal segments. In A23e both the dendrites and axons project more posteriorly in more caudal segments. *Abrupt change:* A18g5 and A18c show abrupt changes in morphology that can only be observed in one segment. A18g5 axon projects to a common region in t1, but it is much shorter in a8. Dashed grey lines indicate the segments boundaries. A18c has two dendritic branches in a1 but only one in all other abdominal segments. The color of the boxes refers to the morphological topology.

c. Schema representing the neuromeres involved in crawls (abdominal) and turns (thoracic and a1-a4).

d. Percentages of morphological topology grouped in categories that better characterized the axonal and dendritic morphology along the anterior-posterior abdominal segments.

Note that not all neurons have a segmental homologue in a8 where both the proliferation and cell death is differentially [36,37] regulated (Supplementary Fig. 1).

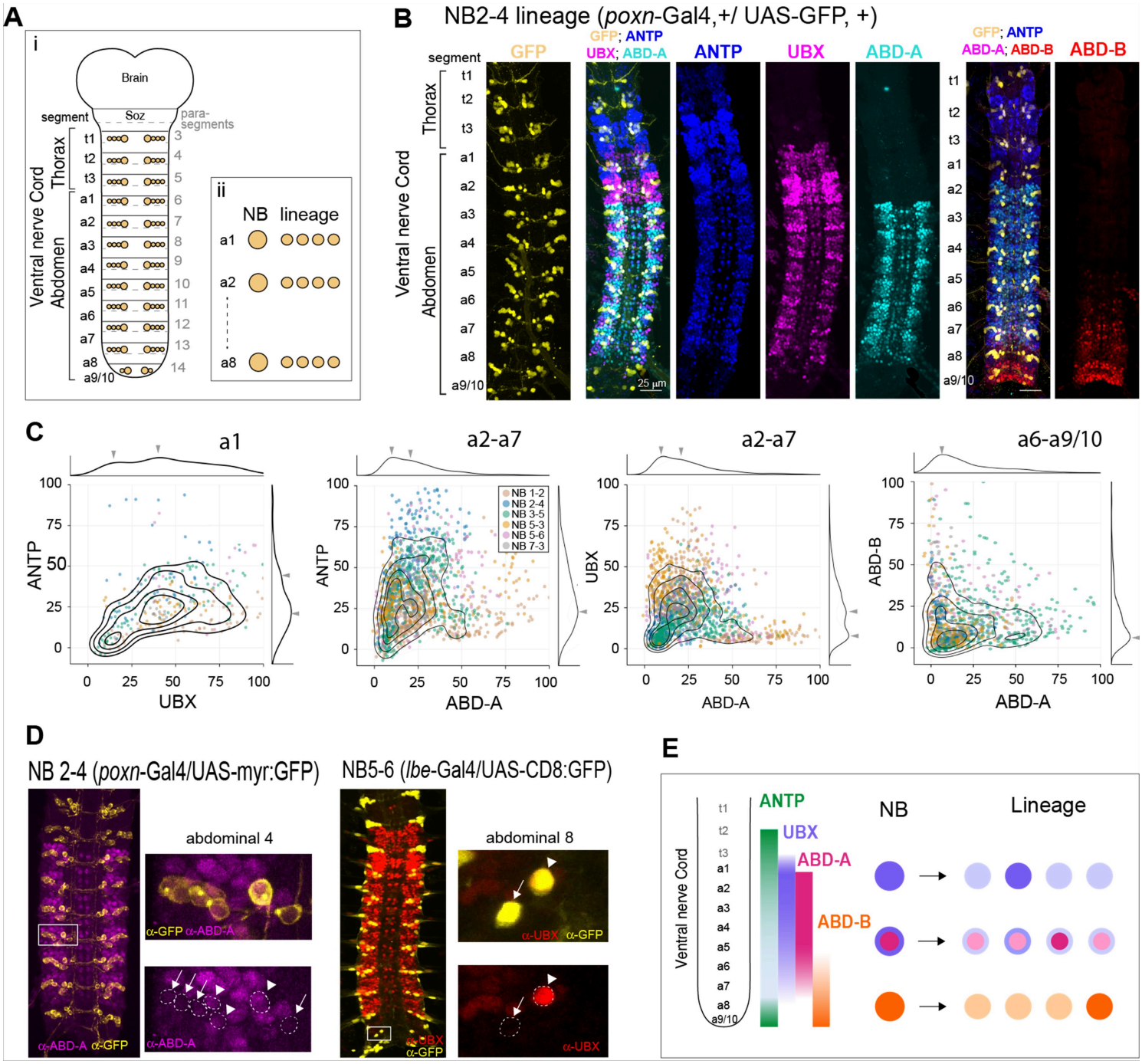


Figure 2

Hox expression is variable in postmitotic neurons along the VNC.

a. Scatter plots showing the levels of Hox protein for each NB lineage in the segments indicated. Hox levels were normalised as percentage of the highest expressing neuron in the same embryo. The left and right side of three individual nervous systems were analysed for each NB studied. Kernel Density Analyses with lines representing 20% density are overimposed. The majority of neurons express the expected generic Hox proteins of the region where they are located, but some neurons have either very low or very high levels of one or several Hox proteins.

b. Dorsal view of a representative stage 16 embryo VNC, showing GFP expression in NB 2-4 lineage stained against ANTP, UBX and ABD-A (left) and against ANTP, ABD-A and ABD-B (right). The pattern of Hox expression is inhomogeneous along the body and in individual segments.

c-d. Examples of sibling interneurons showing different levels of Hox protein even if located in the same segment. **c.** In abdominal 4 of NB2-4 (inset from **b**) three neurons express low levels of the generic ABD-A protein while three other neurons show no expression. **d**. In abdominal 8 of NB 5-6, one interneuron expresses UBX outside its generic domain, while its sister cell does not.

e. Summary schematic showing how different combinations and levels of Hox proteins are present within the same lineage.

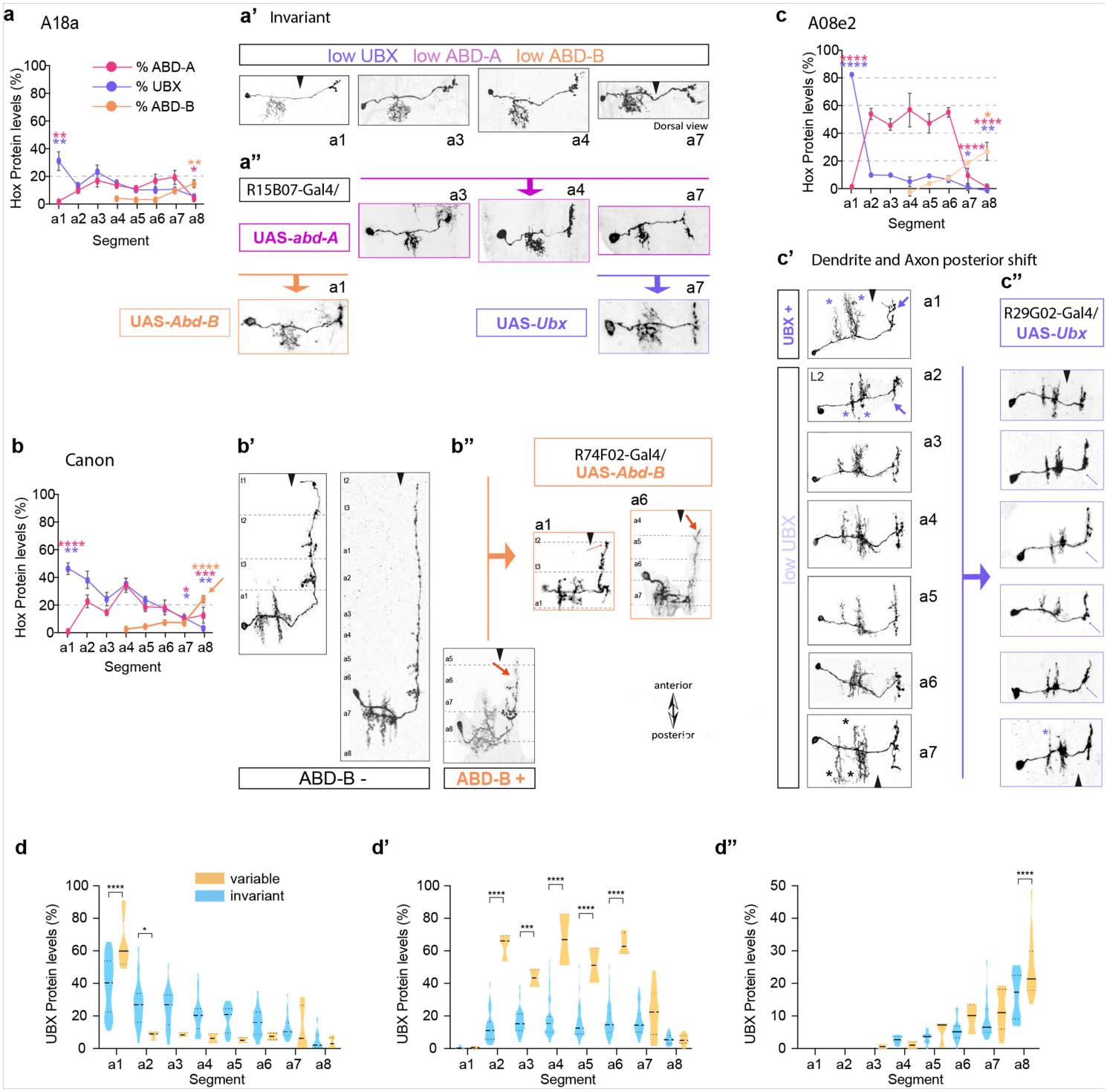


Figure 3

Postmitotic Hox expression controls neuronal morphology.

a-c. Average Hox levels for each abdominal segment. 6<N<16 for each data point. ANOVA with Dunnett's multiple comparisons test was performed for each Hox protein. The values of each segment were compared with the segment a4. * p < 0.05; ** p < 0.01; *** p< 0.001 and **** p<0.0001.

a’-c’. Segmental variants of the neurons quantified. Segment differences in dendrites are indicated by an asterisk while and arrow in axons.

a’’-c’’. Morphological changes when Hox genes are over-expressed. **A’’.** Morphology of A18a is invariant (see Fig. 1). The morphology does not change upon over-expression of *Ubx, abd-A or Abd-B* (n=5 of 5 FlipOuts (FO)). B’’. Over-expressing *Abd-B,* during differentiation, in a1 or a6 transforms the neuronal morphology and the neuron has a short axon as indicated by the arrow (n=3 of 4 FO). **C’’.** Over-expressing *Ubx* induces a unique anterior projection of the axon (n=6 of 12 FO). In a7, *Ubx* over-expression might modify the dendritic distal projection rendering it more anterior (n=1 of 2 FO). The over-expressing neurons were imaged in late first instar larvae to guaranty that the effects are only developmental. The color of the boxes refers to the Hox gene over-expressed. All confocal images were inverted to better reveal the neuronal morphologies.

d-d’’. Percentage of Hox protein levels in variable or invariable neurons. ANOVA with Sídák's multiple comparisons test was performed comparing the two phenotypes in each segment. * $p < 0.05$; *** $p< 0.001$ and **** $p<0.0001$.

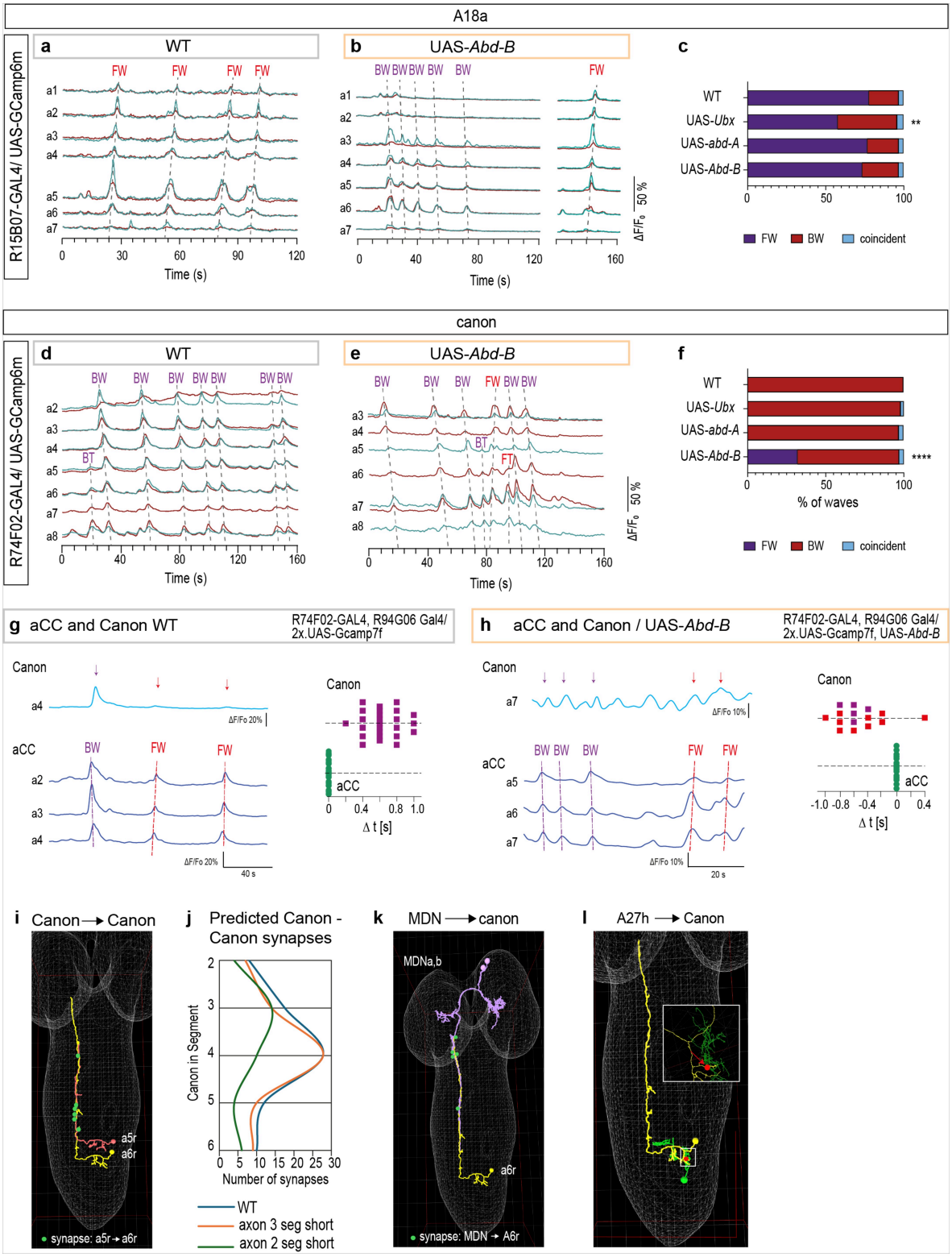


Figure 4

Rewiring changes neurons inclusion into specialized motor network.

a-f. Comparison of representative calcium activity in spontaneous fictive locomotion across ventral nerve cord segments of wildtype and Hox genes over-expressing first instar larvae.

a. A18a wildtype (WT). **b.** A18a > *Abd-B*. **c.** Quantification of activity wave types over-expressing Hox genes in A18a neurons. No novel pattern of activity is observed upon over-expression of *Ubx, abd-A or Abd-B,* however the frequency of wave types is different in A18a > *Ubx* (Fisher exact test comparing WT to *Ubx* = 0.0071).

d. Canon wildtype. **e.** Canon > *Abd-B*. **f.** Quantification activity wave types over-expressing Hoxes in Canon neurons. WT Canon neurons are only active during backward waves. In Canon > *Abd-B* the axon shortened and the patter of activity changes, it is less coordinated and includes forward waves of activity (Fisher exact test comparing WT to *Abd-B,* $p<0.0001$).

A18a WT: 36 waves from six larvae ; A18a > *Abd-B*: 99 waves from five larvae; Canon WT: 44 waves from nine larvae; Canon > *Abd-B*: 37 waves from six larvae.

g,h. Representative examples of simultaneous recording of calcium activity in Canon and the motorneurons aCC. Right panel shows the peak time difference between both neurons. During backward waves, Canon neurons are activated after the motorneuron of the same segment allowing for muscle relaxation after contraction [24]. When in aCC, Canon > *Abd-B* CNSs the waves of activity travels forward, Canon neurons are active before the aCC. The Canon's peak time was also affected during backward waves suggesting a lack of coordination in the active cells. Canon, aCC, Canon WT: 30 BW waves from two larvae; aCC, Canon > *Abd-B*: 15 waves (5 BW, 10 FW) from three larvae.

FW = Forward Wave; FT = Forward Truncated; BW = Backward Wave; BT = Backward Truncated; C= Coincident. Only complete waves were analysed.

i. Location of the synapses, green dot, from Canon a5 onto Canon a6 on the right CNS side.

j. Prediction of the number of synapses received by a Canon neuron in a specific segment in wildtype condition (based on [24]) or if the axon were three segment (seg) short or two segments short.

k. Location of the synapses, green dot, between MDNa and b onto Canon a6 on the right CNS side.

k. Location of the synapse, red dot, from A27h onto Canon a6 on the right CNS side.

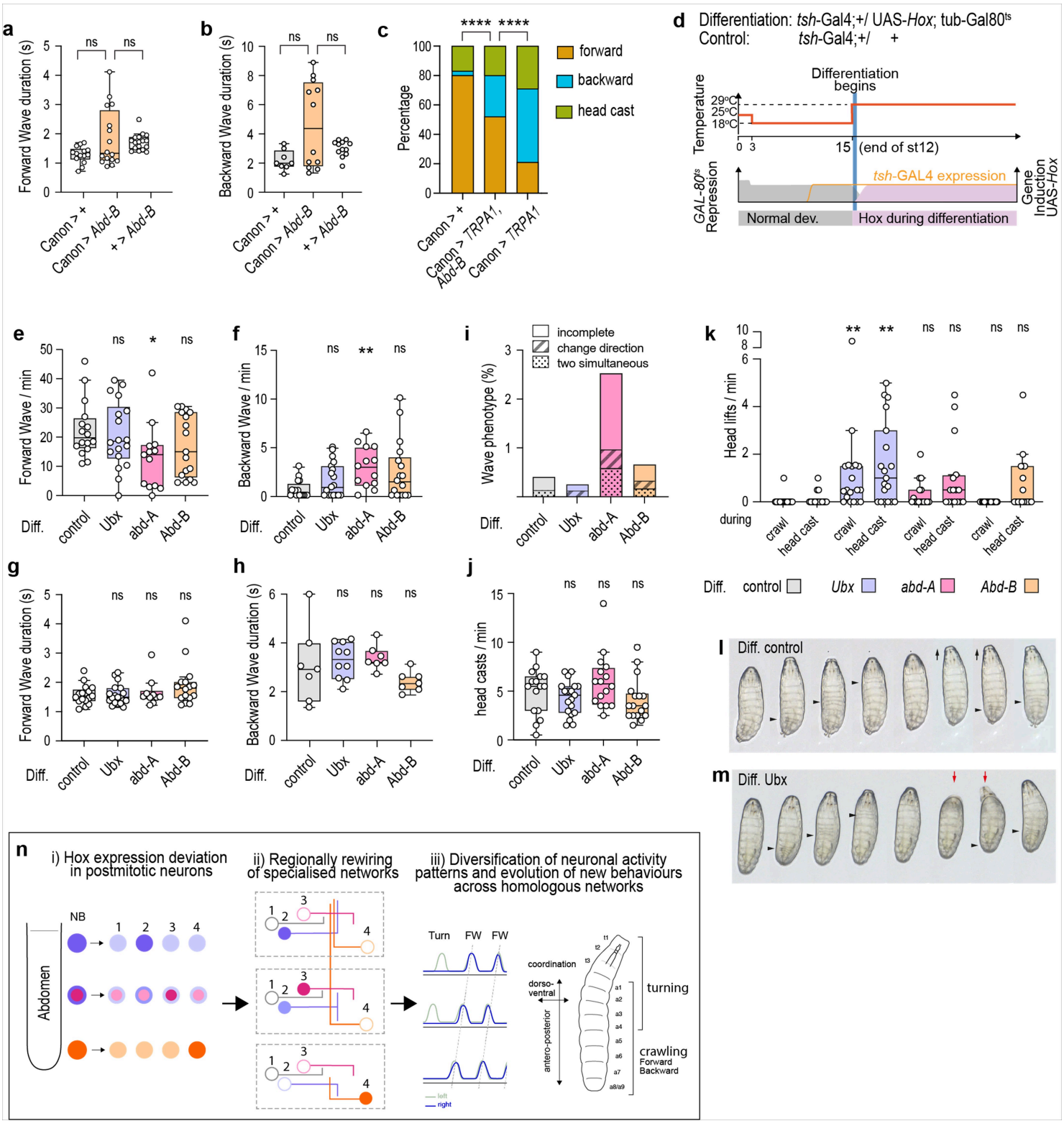


Figure 5

Hox driven rewiring is necessary for the evolution of novel behaviours.

Forward wave duration when over-expressing *Abd-B* in Canon neurons compared to controls.

a. backward wave duration when over-expressing *Abd-B* in Canon neurons compared to controls. **a,b.** Kruskal-Wallis test with Dunn's multiple comparisons test showed no significant differences. Canon > + n= 14; Canon > *Abd-B* n= 16; + > *Abd-B* n= 17.

b. Percentage of forward or backward waves or head casts (including turns) when over-activating Canon neuron and over-expressing *Abd-B* compared to controls. Fisher's exact test on number of events for the three genotypes $p < 0.0001$. Fisher pairwise comparisons **** = $p < 0.0001$.

c. Conditional increase of Hox genes expression during nervous system differentiation (Diff.) was achieved by blocking the VNC Gal4 (*tsh*-Gal4) activity with the temperature sensitive *tub*-Gal80^{ts} at 18°C. Once differentiation started the temperature was increased at 29°C and Hox genes over-expressed.

e-m. Locomotor analysis when increasing Hox genes levels during differentiation.

e. Forward waves per minute. **f.** Backward waves per minute. **g.** Forward waves duration. **h.** backward wave duration. **i.** percentage of wave phenotypes. Fisher's exact test on number of events for the three genotypes $p = 0.98$. **j.** head cast (including turns) per minute. **k.** head listing per minute during either forward crawls or head cast (including turns). **l,m.** Representative sequential pictures of a larva during forward crawl. While the thoracic segments (head) of the control larvae remained touching the substrate during crawls (**l**), the larvae over-expressing *Ubx,* during differentiation, consistently lifted their head (**m**). This newly acquired motor behaviour copies the lifting of abdominal segments necessary for body movement during forward crawling [49]. Head arrow indicate contracted segments, black arrow the head (thorax) extension at the end of a forward wave, while the red arrow shows the head lifting. Kruskal-Wallis test with Dunn's multiple comparisons test were performed comparing the distinct Hox over-expressions with the control. * = $p < 0.05$; ** = $p < 0.01$. Control (*tsh*-Gal4/+; +/+) n = 16 (738 waves); Diff. *Ubx* (*tsh*-Gal4/UAS-*Ubx*; +/ *tub*-Gal80^{ts}) n = 18 (805 waves); Diff. *abd-A* (*tsh*-Gal4/UAS-*abd-A*; +/ *tub*-Gal80^{ts}) n = 13 (520 waves); Diff. *Abd-B* (*tsh*-Gal4/UAS-*Abd-B*; +/ *tub*-Gal80^{ts}) n = 17 (592 waves).

n. Summary of the results reported in this study. i) During differentiation, the expression of Hox genes is variable in homologous interneurons. ii) This variable expression induces segmental cell-type specific differences of neuronal morphology in homologous neurons dictating the assembly of regionally specialized networks. iii) Thus, neurons are integrated into specialized networks inducing the diversification of patterns of activity and behaviors across homologous segments. This process has governed the evolution of new behaviours.

Supplementary Files

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- RodriguezCurtSupplMaterial.pdf
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- NNA95163movie2DiffAbdBtshabdB.mov
- NNA95163movie3DiffUbxtshUbx.mov
- NNA95163movie4DiffabdAtshabdA.mov